JUDGES' LIBRARY / DOSSIER 02

Data & scientific evidence.

Source by source: what is measured, what is calculated, and where interpretation must stop.

Four separate data contexts. Personal readings and synthetic records enter the baseline workflow. NASA archive values support a separate research explorer. NOAA Kp supplies Earth geomagnetic context only.

The source inventory

Source	Used in PROHOR	Do not infer
NASA OSD-575	Submitted Inspiration4 metabolic-panel CSV; research charts and descriptive comparisons.	Live crew telemetry, Mars evidence, or validation of personal alerts.
NASA OSD-569 / OSD-571	Study titles, descriptions and source links.	Additional personal readings or analysis of their laboratory values.
NASA OSD-87	Optional mouse-study metadata, outside the default human filter.	Human clinical thresholds or a user's condition.
NOAA planetary Kp	Earth geomagnetic observation context, with timestamps.	A Mars/cabin reading, individual dose, or cause of a health change.
Local / demo data	User-entered or imported readings; deterministic synthetic demo records.	A live sensor or a NASA clinical dataset.

NASA Human Research Program references provide framing about spaceflight hazards and measurement context. They are not endorsements or validation of PROHOR's numerical rule.

The snapshot and data views label source and freshness. A network fetch time is not the same as the time an observation was measured.

02 / TRACEABLE COMPUTATION

Follow the OSD-575 evidence.

Source, identity and preservation

The explorer reads the original submitted comprehensive metabolic panel for four coded Inspiration4 participants on a three-day 2021 mission. The preserved file contains **531 rows, 19 analytes and seven categorical collection points**: L−92, L−44, L−3, R+1, R+45, R+82 and R+194. L indicates days before launch; R is days after return.

File: LSDS-8_Comprehensive_Metabolic_Panel_CMP.upload_SUBMITTED.csv
Size: 25,275 bytes
SHA-256: dab247eb794edab59aaaa683c85fd1f2dbf4ae3684872739338e10ead4192a0c

The browser parses the source rows and checks the bytes against the bundled manifest where Web Crypto is available. A mismatch blocks archive analysis; lack of a supported checksum is disclosed. A matching hash demonstrates identity relative to the manifest, not scientific correctness or protection against replacement of both file and manifest.

One reproducible result: glucose, mg/dL

Subject	Pre-flight median	R+1 value	R+1 change	R+45 value
C001	80	90	+10	84
C002	72	83	+11	44
C003	75	103	+28	36
C004	83	97	+14	43

Each pre-flight median is calculated from the subject's finite pre-flight observations; the R+1 change subtracts that median. The plotted sample positions are categorical, not evenly spaced days. Non-numeric or censored values are retained as source text and excluded from arithmetic; missing values are not filled.

At R+45, C002, C003 and C004 fall below the recorded laboratory interval (65–99 mg/dL). This is a source-context review cue, not a diagnosis or proof of a flight effect; fasting status and measurement circumstances are not established here.

03 / INTERPRETATION

What the numbers cannot tell us.

Limits before interpretation

Four people and a short mission cannot establish population reference ranges or validate long-duration monitoring. The pre/post pattern cannot isolate mission exposure from other influences. Source laboratory reference intervals are metadata, not PROHOR's personal change bands. Legacy race-labeled eGFR fields are preserved in the raw file but excluded from selectable analyses.

NASA metadata and NOAA Kp

The app's read-only adapter attempts NASA OSDR study metadata and NOAA SWPC planetary Kp through same-origin endpoints. Validated responses may be served from an in-process cache or a bundled fallback when the upstream is unavailable. An observation's timestamp and the retrieval timestamp describe different things. The studies library is curated, not a search across all NASA research.

Data integrity is not clinical validation. Source citation, byte verification and reproducible calculations help a judge check the work. They do not establish clinical sensitivity, specificity, safety or usefulness.

How to reproduce the archive

Download the submitted CSV from the NASA source below, compare the SHA-256, inspect the study metadata and source rows, and run the repository's research tests. The repository's **DATA-PROVENANCE.md** describes source files, normalization and constraints in detail.

Sources & further reading

- NASA OSD-575 study
- Original NASA CSV
- OSD-569
- OSD-571
- NOAA SWPC planetary Kp
- Full data provenance